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virtualis fit realis

Declaration

I hereby declare that this thesis is my own work and that all sources and citations have been properly acknowledged. No part of this thesis has been submitted for any other degree or professional qualification at this or any other institution.

Abstract

The abstract should briefly describe the context of the research, the central research question, the methodology used, the most important results, and the main conclusion. Keep it concise and informative, aiming to give readers a clear overview of the entire work.

Acknowledgements

I would like to thank [Name] for [their] guidance and support throughout the course of this thesis. I also wish to express my gratitude to [institution/lab] for providing the resources necessary to carry out this research, and to my family and friends for their encouragement.

Abbreviations

SoE Sense of Embodiment

SoO Sense of Ownership (also known as sense of body ownership, SoBO)

SoA Sense of Agency

SoL Sense of Self-Location

VR Virtual Reality

XR Extended Reality

IK Inverse Kinematics

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Chapter 1

Introduction

1.1 Motivation

Indoor rowing is a physically demanding and technically structured form of exercise. Unlike many stationary fitness activities, rowing involves a coordinated whole-body movement in which legs, trunk, arms, seat, handle, and rhythm are tightly coupled. On a rowing ergometer, this movement is reduced to a land-based training device: the user moves back and forth on a sliding seat and pulls a handle along a mostly linear path. The physical effort can approximate important aspects of rowing training, but the perceptual situation differs strongly from rowing on water. The user remains stationary in a room, the boat is absent, and the characteristic visual and spatial experience of moving through a rowing environment is largely missing.

Virtual reality offers an opportunity to reintroduce some of this missing experiential context. A virtual rowing system can place the user in a boat, on a river, or in a training environment that visually resembles the sport more closely than an ergometer display can. Such a system can add optic flow, spatial sound, avatar representation, opponents, teammates, and environmental feedback. In stationary exercise contexts, this visual motion through a virtual environment can also contribute to an impression of self-motion, even when the user's physical body remains in place.

However, this opportunity also introduces a design problem. The ergometer makes rowing available indoors by replacing the boat and oars with a machine. Virtual reality can visually restore the boat, oars, water, and motion through space, but it cannot remove the fact that the user is still physically rowing on an ergometer. The system must therefore decide how the real movement should become a virtual rowing action.

This decision matters because indoor rowing in VR is not simply a ques-

tion of adding a more attractive background to exercise. The user performs a real, repetitive, forceful movement while seeing a mediated version of that movement in a virtual environment. The virtual scene can make the activity feel more like rowing on water, but the physical interface remains the ergometer. As a result, the system must balance two different forms of fidelity: fidelity to the user’s actual sensorimotor interaction with the machine, and fidelity to the visual and cultural form of the sport.

This thesis investigates that balance. It examines VR indoor rowing as a case where a physical exercise device stands in for a more complex sport action. The central question is whether the virtual representation should preserve the user’s actual interaction with the ergometer as directly as possible, or transform that interaction into the boat- and oar-based action associated with rowing on water.

1.2 The Core Design Tension

A rowing ergometer and a rowing boat do not provide the same movement interface. On an ergometer, the user sits on a sliding seat and pulls a handle along a constrained path. In a rowing boat, the rower interacts with oars that rotate around oarlocks, blades that interact with water, and a boat that moves relative to the environment. Even when the training movement is related, the physical apparatus and spatial logic are different.

A VR rowing system can respond to this difference in at least two plausible ways. In an ergometer-congruent representation, the virtual body, handle, and rowing apparatus are aligned primarily with the user’s real machine and movement. The visible hands and handle can remain close to where the user feels them, and the avatar can be animated around the actual handle path and seat motion. This preserves the physical logic of the user’s current interaction.

In a boat-centric representation, the system instead prioritizes the visual and sport-specific logic of rowing on water. The user may appear to sit in a boat, move through a river, and row with oars. This can make the virtual scene more recognizable as rowing, but it may require transforming the user’s real handle movement into a virtual oar movement that does not fully coincide with the felt motion of the hands and handle.

The comparison is therefore not between a real and an artificial representation of rowing. Both representations are artificial in different ways. The ergometer-congruent condition preserves the physical logic of the user’s actual device and movement, but may depict an indoor machine as if it were moving through a virtual rowing environment. The boat-centric condition

preserves the visual and cultural logic of the sport, but must transform the user’s straight ergometer handle movement into a boat-and-oar action. There is no neutral third option in which the indoor ergometer becomes on-water rowing without remainder.

This makes VR indoor rowing a useful case for studying a broader XR design problem. Many XR sport and exercise systems use stationary or simplified physical equipment to represent more complex activities. The system can either make the virtual action correspond closely to the user’s real movement, or it can transform the movement to better match the represented activity. Both choices solve one part of the problem while introducing another.

In this thesis, this design problem is framed as a trade-off between visuomotor congruence and realistic sport depiction. Visuomotor congruence refers to the correspondence between the user’s performed movement and the movement they see in the virtual environment. Realistic sport depiction refers to the extent to which the virtual scene and action resemble the sport being represented. The two can support each other, but in indoor rowing they can also come apart.

1.3 Embodiment and Experiential Evaluation

The trade-off between ergometer-congruent and boat-centric representation is especially relevant for virtual embodiment. Virtual embodiment concerns the extent to which users experience a virtual body or body part as related to their own body. Commonly discussed components include the sense of agency over the virtual body’s actions, the sense of body ownership, and self-location in relation to the virtual body or space.

The ergometer-congruent condition is closely related to principles from virtual hand and body ownership research. In such paradigms, embodiment can be influenced by whether seen and felt body positions, movements, or touches are spatially and temporally aligned. In the rowing case, the real and virtual hands or handle can potentially be placed in close correspondence: the user feels the physical handle and sees the virtual hands or handle move in the same way.

However, rowing also differs from classic hand-ownership paradigms. It is an active, rhythmic, tool-mediated, whole-body exercise rather than a passive illusion involving a single limb. The user does not merely observe a virtual hand; they exert force, coordinate multiple body segments, and interact with a physical machine. This makes indoor rowing an interesting extension of embodiment questions into a sport and exercise context.

The boat-centric condition may be less directly congruent with the user’s felt hand and handle movement, but it may better express what users understand as rowing. A virtual boat, oars, water, and forward movement can contribute to perceived sport realism and plausibility. For a rower or rowing-interested user, this representation may feel more meaningful or more appropriate to the activity, even if the mapping between real and virtual movement is less direct.

For this reason, the thesis does not treat embodiment as the only relevant outcome. The primary focus remains on embodiment-related experience, particularly agency, ownership, self-location, and overall sense of embodiment. However, the study also considers perceived rowing realism and user preference as design-facing measures. A representation can be sensorimotor-congruent but still be experienced as a poor depiction of rowing; conversely, a representation can be less congruent but preferred because it better captures the intended sport scenario.

Preference is therefore treated as an exploratory complement rather than as the main theoretical construct. It is included because practical XR sport systems are not evaluated only by whether they produce embodiment, but also by whether users find the representation convincing, appropriate, and desirable for the activity. At the same time, preference must be interpreted cautiously, because it may depend on visual polish, prior rowing experience, individual expectations, and details of the implementation.

1.4 Research Gap

Research on virtual embodiment has shown that multisensory congruence, visuomotor synchrony, and avatar representation can influence how users experience virtual bodies and actions.

A separate body of work has explored virtual reality and interactive systems for sport, exercise, and rowing. These systems often aim to support training, motivation, feedback, or presence by augmenting stationary exercise with virtual environments, avatars, or simulated sport contexts.

Existing rowing-related VR and simulation work already contains many of the ingredients relevant to this thesis. Some systems use ergometer like vehicles as virtual representations. Others use 3D models of rowing machines or rowing machine inspired vehicles to represent the User. Some systems use inverse kinematics or avatar animation to represent the user’s rowing movement. Commercial and consumer-facing applications also illustrate different representational choices, ranging from travelling through virtual environments on an indoor machine to more boat- or oar-centered rowing depictions.

However, the specific comparison addressed in this thesis appears to be missing from the reviewed literature. Existing studies tend to compare VR against non-VR rowing, evaluate feedback or visualization techniques, manipulate avatar properties, adjust simulator parameters, or study motivational and social factors. They generally do not isolate the physical-to-virtual rowing mapping itself as the central experimental variable.

More specifically, the reviewed material does not appear to contain a controlled study that holds the physical rowing setup constant while comparing an ergometer-based mapping against a boat-based mapping as the main independent variable. Such a comparison would ask whether the avatar and rowing apparatus should be solved primarily in the coordinate frame of the real ergometer, or transformed into the coordinate frame of a virtual boat and oar system.

This gap is important because the mapping choice is not only a technical implementation detail. It changes the relation between the user’s body, the physical machine, and the virtual action they see. As a result, it may influence agency, body ownership, self-location, perceived realism, and preference. If the mapping is not isolated experimentally, it remains unclear whether a more sport-realistic rowing depiction supports the experience more strongly than a more sensorimotor-congruent ergometer depiction, or whether the opposite is the case.

The thesis addresses this gap by focusing on the mapping choice itself. It compares an ergometer-congruent representation and a boat-centric representation within the same general VR rowing context. The aim is not to evaluate every aspect of VR rowing, but to examine how this specific design choice shapes embodied and experiential responses.

1.5 Research Objectives and Questions

The objective of this thesis is to investigate how different representation strategies in VR indoor rowing influence embodiment-related and experiential outcomes. Specifically, the thesis compares an ergometer-congruent representation with a boat-centric representation. The comparison is guided by the assumption that both strategies can be justified by different design goals: the ergometer-congruent mode may support sensorimotor alignment and agency, while the boat-centric mode may support perceived rowing realism and the impression of rowing on water.

The first objective is to design and implement a VR rowing prototype that can represent the same physical ergometer activity in two different ways. In the ergometer-congruent mode, the system emphasizes direct correspondence

between the user’s real movement and the virtual body or handle. In the boat-centric mode, the system emphasizes a rowing scene with boat and oar depiction, even if this requires transforming the user’s handle movement into a more sport-like visual movement.

The second objective is to use this prototype as a research instrument. The system is not only an application prototype, but also a means of examining how physical-to-virtual movement mapping affects user experience. This requires the two modes to be comparable in as many respects as possible, so that the central difference remains the relation between the real ergometer movement and the virtual rowing depiction.

The third objective is to evaluate the two representation modes with respect to embodiment-related constructs. These include sense of agency, body ownership, self-location, and overall sense of embodiment. In addition, the study considers perceived rowing realism, comfort, and preference as exploratory design-facing outcomes.

The thesis is guided by the following main research question:

How does the trade-off between visuomotor congruence and realistic sport depiction affect embodiment, perceived rowing realism, and user preference in VR indoor rowing?

This main question is divided into the following subquestions:

1. How can an ergometer-congruent and a boat-centric rowing representation be implemented while keeping the physical task and virtual rowing context broadly comparable?
2. How do the two representation modes differ in embodiment-related experience, particularly sense of agency, body ownership, self-location, and overall sense of embodiment?
3. How do the two representation modes differ in perceived rowing realism, comfort, and user preference?
4. What do the results suggest for the design of XR sport and exercise systems in which physical equipment and virtual sport actions do not fully coincide?

These questions are intentionally framed as comparative and exploratory rather than as claims of expected superiority. It is plausible that the ergometer-congruent mode supports stronger agency because the visible movement more closely matches the user’s physical action. It is also plausible that the boat-centric mode supports a stronger sense of rowing realism because it better

represents the visual and cultural form of the sport. The study therefore examines how these dimensions interact rather than assuming that one form of fidelity is always preferable.

1.6 Contributions

This thesis makes three main contributions.

First, it presents the design and implementation of a VR indoor rowing prototype that supports two distinct representation strategies for the same physical rowing activity. The prototype distinguishes between an ergometer-congruent mode and a boat-centric mode. This distinction makes it possible to examine a design trade-off that is often implicit in XR sport systems: whether virtual representation should prioritize the user’s actual input apparatus and body movement, or the visual form of the sport being simulated.

Second, the thesis contributes an embodiment-oriented comparison of these two representation strategies. Rather than treating immersion, realism, or motivation as broad and undifferentiated goals, the thesis focuses on how movement mapping may affect specific components of embodied experience. In particular, it considers sense of agency, body ownership, self-location, and overall sense of embodiment as relevant constructs for evaluating rowing representation in VR.

Third, the thesis provides design implications for researchers and developers of XR sport and exercise systems in which physical equipment and virtual sport depiction do not fully coincide. The rowing ergometer is a clear example of this situation, but the underlying problem also applies to other stationary training systems where real input devices are used to represent more complex sport actions. The thesis therefore contributes to a broader discussion about how embodied XR systems should balance sensorimotor congruence, visual plausibility, sport-specific realism, and user preference.

The intended contribution is not a general claim that one representation strategy is universally superior. Instead, the thesis aims to clarify what is gained and what may be lost when a VR sport system chooses between direct physical congruence and realistic sport depiction. Both strategies solve part of the design problem while introducing another. A congruent representation may feel controllable and bodily grounded, while a sport-realistic representation may feel more meaningful or recognizable as rowing. Understanding this trade-off can help future systems make more deliberate representation choices.

In practical terms, the thesis also documents the technical and methodological considerations involved in building such a comparison. These include

alignment between physical and virtual rowing spaces, avatar motion mapping, control of visual confounds, and the selection of measures for evaluating embodiment and experience. These details are part of the contribution because the comparison depends not only on the final visual result, but on the system’s ability to isolate the relevant design difference.

1.7 Thesis Structure

The remainder of this thesis is structured as follows.

Chapter 2 introduces the theoretical and empirical background for the work. It discusses virtual embodiment, sense of agency, body ownership, self-location, visuomotor congruence, and related concepts such as presence and vection. It also reviews relevant work on VR exercise, sport simulation, and rowing-related systems. The chapter establishes the conceptual basis for treating movement mapping as a central design factor in VR rowing.

Chapter ?? describes the VR rowing prototype developed for this thesis. It presents the hardware and software setup, the alignment and tracking approach, and the two representation modes. The ergometer-congruent mode and boat-centric mode are described in terms of their mapping strategies, avatar motion, and intended experiential differences. The chapter also discusses implementation constraints and design decisions that affect the comparison.

Chapter 4 presents the study design used to evaluate the two representation modes. It describes the experimental conditions, participant procedure, measures, and analysis approach. Particular attention is given to how embodiment-related constructs, perceived rowing realism, comfort, and preference are measured. The chapter also explains how the comparison is structured to reduce confounds between the two modes.

Chapter 5 reports the results of the evaluation. Quantitative and qualitative findings are presented with respect to the research questions, including measures related to agency, ownership, self-location, embodiment, perceived realism, comfort, and preference where applicable. The chapter focuses on reporting the observed outcomes without extensive interpretation.

Chapter 6 interprets the findings in relation to the theoretical background and research questions. It discusses whether the results support the importance of visuomotor congruence, sport-realistic depiction, or a more nuanced relationship between the two. The chapter also considers limitations of the prototype and study design, as well as implications for future XR sport systems.

Chapter 7 concludes the thesis by summarizing the main findings and

contributions. It revisits the central trade-off between ergometer-congruent and boat-centric representation and outlines directions for future work in VR rowing, embodied interaction, and XR sport design.

Chapter 2

Theoretical Background and Related Work

2.1 Virtual Embodiment

Virtual reality can do more than surround the user with a simulated environment. It can also provide the user with a virtual body, body parts, tools, or task-specific apparatus that appear to occupy the space of action. In such cases, the user does not only perceive a scene from the outside, but may experience a relation between their own body and the body or action represented in the virtual environment. This relation is commonly discussed under the concept of virtual embodiment.

Virtual embodiment is usually not treated as a single, indivisible feeling. Instead, it is often described through multiple related components, especially body ownership, sense of agency, and self-location. Body ownership concerns the experience that a body or body part belongs to oneself. Sense of agency concerns the experience of causing or controlling an action. Self-location concerns where the user experiences themselves to be located in relation to a body or space.

This distinction is important because the components of embodiment can diverge. A user may feel that they control a virtual hand because it moves in response to their own action, while still not strongly feeling that the hand is part of their body. Conversely, a virtual body may appear spatially aligned with the user and therefore support self-location, while delayed or mismatched movement may reduce agency. Embodiment is therefore better understood as a structured experience in which different cues can support different aspects of the relation between the physical and virtual body.

Many embodiment studies build on the logic of body illusion paradigms.

In the rubber hand illusion, for example, synchronous visual and tactile stimulation can lead participants to report ownership over an artificial hand. Virtual hand and virtual body studies extend this logic into immersive environments, where the virtual body can respond to the user’s movement and may be experienced as part of the user’s body representation.

For this thesis, the most relevant aspect of virtual embodiment is not the mere presence of a virtual avatar, but the relation between the user’s real movement and the movement shown in VR. In indoor rowing, the user performs a real, physically effortful action on an ergometer. If a virtual body or rowing apparatus is shown, the system must decide how this physical action is represented. The visible movement may closely correspond to the user’s actual handle and body motion, or it may be transformed into a visually more sport-specific boat and oar action.

This makes VR rowing a useful case for examining embodiment in a tool-mediated and exercise-based context. The user is not only looking at a virtual body, nor only moving a hand in empty space. They are seated on a physical machine, holding a real handle, pushing against foot stretchers, and performing a rhythmic full-body movement. The virtual representation must therefore mediate between the felt physical apparatus and the visual action shown in the virtual environment.

The central concern of this thesis is how this mediation affects embodiment-related experience. An ergometer-congruent representation may support embodiment because the seen action can remain close to the user’s felt movement. A boat-centric representation may support the plausibility and meaning of the sport scenario, but may require a greater transformation between physical action and visual movement. The following sections therefore examine the components of embodiment and the role of visuomotor congruence in more detail.

2.2 Agency, Ownership, and Self-Location

The sense of embodiment is commonly discussed through three partially separable components: sense of agency, body ownership, and self-location. These components are useful for this thesis because the two rowing representations may not affect all aspects of embodiment in the same way. A representation that supports control may not necessarily feel most like rowing, and a representation that looks most like rowing may not necessarily provide the strongest correspondence with the user’s felt body.

Sense of agency refers to the experience of causing or controlling an action. In virtual environments, agency is often supported when a virtual body or

object moves in a way that matches the user’s own intentions and movements. When a user moves their hand and immediately sees a virtual hand perform the same movement, the visual feedback can support the impression that the virtual action is their own. Conversely, delays, spatial offsets, or unexpected transformations can weaken the sense that the observed action is under direct control.

For VR rowing, agency is especially important because the activity is active, continuous, and rhythmic. The user repeatedly initiates and controls a stroke cycle. If the virtual hands, handle, or rowing body visibly follow the user’s real movement, agency may be reinforced throughout the stroke. If the virtual action transforms the user’s movement into a different oar or boat motion, agency may still be present, but it may depend on whether the transformation feels predictable, responsive, and meaningful.

Body ownership refers to the experience that a body or body part is one’s own. Ownership has often been studied through hand and body illusions, where visual, tactile, proprioceptive, and motor cues are combined. Synchronous and spatially plausible cues can support ownership, while incongruent cues can reduce it. In VR, ownership may be influenced by the visual appearance of the avatar, the spatial alignment between the physical and virtual body, and the correspondence between real and virtual movement.

In the context of rowing, body ownership is relevant because the user may see a virtual body performing the rowing action. An ergometer-congruent representation may place the virtual hands and body in closer correspondence with the felt body and handle. This could support ownership because the virtual body appears to occupy a plausible relation to the user’s own body. A boat-centric representation may show a body engaged in a more recognizable rowing action, but if the visual hand or arm movement diverges from the felt handle movement, ownership may be affected differently.

Self-location concerns the experienced location of the self in relation to a body or environment. In immersive VR, the user’s viewpoint is usually placed inside or near a virtual body, which can support the impression of being located in the virtual scene. Self-location may be influenced by visual perspective, body alignment, movement, and multisensory cues.

Self-location is relevant to VR rowing because the user’s physical body is stationary in a room, while the virtual scene may depict movement through water or a rowing environment. The user may feel located on the ergometer, in the virtual boat, or in a hybrid relation between both. In an ergometer-congruent mode, the felt machine and visible apparatus may reinforce the physical setup. In a boat-centric mode, the visual environment may encourage the user to interpret themselves as being in a boat, even though the physical cues still come from the ergometer.

The three components of embodiment therefore provide different lenses on the same mapping problem. Agency emphasizes the relation between intention, movement, and feedback. Ownership emphasizes whether the virtual body or body parts are experienced as belonging to the user. Self-location emphasizes where the user experiences themselves to be situated. A central assumption of this thesis is that the ergometer-congruent and boat-centric representations may produce different patterns across these components rather than a single uniform increase or decrease in embodiment.

2.3 Visuomotor Congruence and Synchrony

Visuomotor congruence describes the correspondence between a user’s physical movement and the movement they see. In virtual embodiment research, this correspondence is one of the central cues linking the physical body to the virtual body. When the user performs an action and the virtual body moves in a spatially and temporally corresponding way, the virtual movement can be interpreted as the visual consequence of the user’s own action.

Two aspects of congruence are especially relevant: temporal synchrony and spatial correspondence. Temporal synchrony concerns whether the virtual movement occurs at the same time as the user’s real movement. Even small delays can affect the perceived connection between action and feedback, particularly for agency. Spatial correspondence concerns whether the virtual movement occurs in the expected position, direction, and configuration relative to the user’s body. A movement can be temporally synchronous but spatially transformed, or spatially aligned but delayed.

The distinction between temporal and spatial congruence matters for VR rowing. A boat-centric representation may respond immediately to the user’s stroke, preserving temporal synchrony, while still transforming the spatial path of the hands into an oar-based movement. An ergometer-congruent representation may preserve both the timing and the approximate spatial logic of the handle pull, because the virtual hands and handle can follow a path closer to the real ergometer movement. Both representations may therefore be responsive, but they differ in how much of the user’s physical movement geometry is preserved.

Classic body illusion research often emphasizes the importance of synchronous multisensory stimulation. In the rubber hand illusion, synchronous visual and tactile cues can support ownership over an artificial hand, while asynchronous cues tend to weaken the illusion. In active virtual hand paradigms, the relationship between the user’s own movement and the seen movement becomes especially important, because movement itself provides a continuous

cue linking the physical and virtual body.

In VR rowing, the user’s hand movement is not isolated from the rest of the body. The stroke is a coordinated sequence involving leg drive, trunk swing, arm pull, handle movement, and seat motion. This makes congruence more complex than simply matching a virtual hand to a real hand. A system can align some cues while transforming others. For example, the timing of the stroke may match the user, while the visible trajectory of the virtual hands may follow an oar-based path rather than the real ergometer handle path. Conversely, an ergometer-congruent representation can align the hand and handle more closely while sacrificing some of the visual realism associated with rowing on water.

This is the core reason why the comparison in this thesis is not a simple test of realism. A realistic-looking boat scene may provide strong contextual plausibility, but still introduce a visuomotor transformation between the felt and seen action. A visually less sport-realistic ergometer representation may provide stronger congruence with the user’s physical action. The two representation strategies therefore emphasize different forms of fidelity.

The expected role of visuomotor congruence is especially strong for sense of agency. If the virtual action closely follows the user’s intended and performed movement, users may experience the action as more directly their own. Body ownership may also benefit from congruence, especially when the virtual hands and body appear spatially aligned with the felt body. Self-location may be affected more indirectly, depending on whether the user experiences themselves as situated in the virtual body, the physical ergometer setup, or the represented rowing environment.

At the same time, congruence should not be treated as the only value in the design of VR sport systems. A perfectly congruent representation of the ergometer may not be the representation users find most meaningful as rowing. A transformed boat-centric representation may reduce spatial congruence while increasing perceived sport realism. This thesis therefore examines visuomotor congruence as a central mechanism, but not as the only determinant of the overall experience.

2.4 Virtual Hands, Tools, and Body Mapping

In many virtual embodiment studies, the virtual hand or body is treated as the primary object of embodiment. For rowing, however, the user’s action is not only body-centred. It is also mediated through a physical handle, a seat, foot stretchers, and, in the boat-centred representation, virtual oars. The user’s movement is therefore distributed across body parts and tools. This

makes the mapping problem more complex than simply placing a virtual hand where the real hand is.

Tools can become closely integrated into action. A rower does not usually experience the handle or oar as a detached object that must be consciously controlled at every moment. Instead, skilled movement often treats the tool as part of the action system. In VR, this matters because the system can choose whether to represent the felt tool directly, as in an ergometer handle, or to transform it into a sport-specific tool, as in a virtual oar.

This distinction is central for the two conditions in this thesis. In the ergometer-congruent mode, the visible apparatus can remain close to the physical one. The user pulls an ergometer handle and sees a virtual handle and body movement that follow the same basic logic. The virtual representation is therefore anchored in the physical apparatus. This may support embodiment because visual, proprioceptive, and haptic cues point toward the same object relation.

In the boat-centred mode, the same physical action is interpreted as rowing a boat. The user still holds a real ergometer handle, but the virtual representation may show hands, oars, blades, and a moving shell. This is more meaningful as a depiction of the sport, but it introduces a representational translation. The felt handle and the seen oar are not identical tools. The system must therefore preserve enough timing and controllability for the user to accept the mapping, while allowing the visual scene to behave like rowing on water.

Virtual hands are especially important in this translation. Hands are visually salient, close to the user's body, and directly involved in the stroke. If virtual hands are spatially aligned with the felt hands, they can strengthen the link between the physical and virtual body. If they are moved into positions required by the oar animation, they may look more correct as rowing hands, but less directly matched to the felt handle path.

The body mapping problem extends beyond the hands. Rowing is a coordinated full-body action. The legs, trunk, arms, and seat movement together define the stroke. A system that tracks only the handle may reproduce the rhythm of rowing but miss important bodily cues. A system that also estimates seat position and body posture can create a more coherent avatar, but still has to decide whether that avatar should follow the ergometer action or the boat action.

For this reason, this thesis treats body mapping as an embodied design variable rather than only an implementation detail. The same physical stroke can be mapped with different priorities. One mapping can privilege correspondence with the user's felt motion. Another can privilege the recognizable structure of on-water rowing. The question is not simply which mapping is

more realistic in a visual sense, but which mapping better supports the user's experience of agency, ownership, and self-location.

2.5 Presence, Vection, and Stationary Self-Motion

Embodiment is related to, but not identical with, presence. Presence concerns the experience of being in a mediated environment, while embodiment concerns the experienced relation between the self and a body or body part. In VR, these experiences often interact. A believable environment may make the virtual body feel more situated, and a coherent body may make the environment feel more actionable. Nevertheless, the concepts should remain separate because a user can feel present in a scene without strongly experiencing a virtual body as their own.

This distinction is important for VR rowing because the boat-centred mode may increase the plausibility of the sport scenario. Seeing water, a shell, oars, and forward movement may make the situation feel more like rowing. This does not automatically mean that embodiment is stronger. A scene can be convincing as a rowing environment while the virtual hands or body still feel less directly connected to the user's own movement.

The visual experience of moving through space is also relevant. In stationary exercise systems, the body remains in one physical location while the virtual environment may depict locomotion. Optic flow can create a visual impression of self-motion, and vection describes the illusion of self-motion in a stationary observer. In rowing, this can make an ergometer action feel connected to movement through a river or race course, even though the user remains in the room.

For this thesis, vection is not the main dependent construct, but it helps explain why a boat-centred representation may feel meaningful. The user's effort produces visible locomotion. The stroke is therefore not only an isolated body movement, but appears to move the user through a virtual environment. This can support plausibility and motivation, and may also influence self-location by encouraging the user to experience themselves as situated in the virtual rowing scene.

At the same time, stationary self-motion is a design tension. Visual motion that is not sufficiently related to the user's action can create a mismatch between visual and bodily cues. In many VR applications, such mismatches are discussed in relation to cybersickness. In rowing, the risk may be moderated by the fact that visible motion is coupled to the user's own rhythmic

effort rather than imposed independently. However, this should be treated cautiously and empirically rather than assumed.

The two rowing representations therefore differ not only in body mapping, but also in how strongly they situate the user in a moving sport scenario. The ergometer-congruent mode may offer stronger correspondence with the physical body and machine. The boat-centred mode may offer stronger environmental plausibility and visually supported self-motion. This makes presence andvection useful background concepts, while the thesis remains focused on embodiment and its subcomponents.

2.6 Rowing Biomechanics and Ergometer Rowing

Rowing is a cyclical, whole-body movement that is commonly described through phases such as the catch, drive, finish, and recovery. During the drive, the rower pushes with the legs, transfers force through the trunk, and completes the stroke with the arms. During the recovery, the sequence is reversed as the rower moves back toward the catch. This rhythm is central to both ergometer rowing and on-water rowing, but the two are not mechanically identical.

An ergometer constrains the rowing action through a fixed machine frame, handle path, sliding seat, and foot stretchers. It allows the user to perform a rowing-like movement indoors and to generate measurable work, but it does not reproduce all forces and spatial relations of a moving boat. In on-water rowing, the rower interacts with oars, blades, riggers, boat movement, and water resistance. The hands and oars follow a rowing-specific action that includes blade handling, squaring and feathering, and the relation between the shell and the water.

This difference is important for the present thesis because the user physically rows on an ergometer while the virtual environment can depict either an ergometer-centred action or an on-water rowing action. A direct one-to-one mapping from physical movement to virtual boat rowing is therefore not possible in a strict biomechanical sense. The system must choose which aspects of the stroke to preserve and which aspects to transform.

The ergometer-congruent representation preserves the indoor rowing apparatus as the main frame of reference. It can align the virtual handle, seat, and body movement with the user's felt setup. This makes the virtual action less like a full on-water rowing scene, but more directly connected to what the user is physically doing.

The boat-centred representation preserves the cultural and sport-specific image of rowing. It can show the user as sitting in a shell, moving through water, and interacting with oars. This may better match what users imagine when they think of rowing as a sport. However, the visual hand and oar movements may need to diverge from the exact physical handle trajectory of the ergometer.

This creates a practical form of fidelity trade-off. Biomechanical or visuomotor fidelity to the ergometer is not the same as contextual fidelity to rowing on water. A system can be faithful to the user’s physical input while looking less like the sport, or faithful to the sport depiction while transforming the user’s physical input. This trade-off provides the concrete movement basis for the embodiment comparison in this thesis.

2.7 VR Sport and Rowing-Related Work

VR sport systems combine bodily action with simulated environments. They can be used for training, exercise, rehabilitation, motivation, or entertainment. Compared with passive VR experiences, sport and exercise systems place greater demands on the relation between movement, feedback, and physical effort. The user is not only observing a scene, but acting continuously within it.

A recurring issue in VR sport is fidelity. Fidelity can refer to the visual realism of the environment, the realism of the task, the correspondence between real and virtual movement, or the usefulness of the simulation for training transfer. These forms of fidelity are related, but they are not the same. A system can look realistic while simplifying movement, or reproduce movement constraints while presenting a more abstract environment.

Rowing has appeared in several virtual or interactive training contexts. Some work has investigated VR or virtual environments for rowing exercise, training, social presence, or transfer between virtual and real rowing. These studies are useful as related work, but the present thesis addresses a more specific embodiment question. It does not primarily evaluate training effectiveness, physiological performance, or long-term motivation. Instead, it asks how two representation strategies affect the experienced relation between the user’s body, movement, and virtual rowing action.

Adjacent exercise systems such as VR cycling are also relevant because they show how stationary physical effort can be connected to virtual locomotion. In such systems, visual motion, environmental context, avatars, and interaction feedback can influence the exercise experience. However, rowing differs from cycling because the hands, handle, seat, trunk, and legs form

a highly visible coordinated movement. The user’s upper body and tool interaction are therefore more central to the embodied visual representation.

The contribution of this thesis is therefore not simply another VR rowing prototype. Its focus is the comparison between an ergometer-centred, high-congruence representation and a boat-centred, sport-depiction representation within the same general VR rowing context. This comparison separates two design values that are often blended together: close correspondence with the user’s actual movement and convincing depiction of the sport scenario.

Related work in VR sport provides the broader design context, while embodiment research provides the conceptual lens. Rowing biomechanics explains why the two mappings are not equivalent. Together, these perspectives motivate the empirical question: when the user rows on an ergometer in VR, is embodiment better supported by preserving the felt movement geometry or by presenting a more recognizable on-water rowing action?

2.8 Synthesis for This Thesis

The theoretical background can be summarized as a mapping problem. Virtual embodiment depends on how the user’s physical body, actions, and sensory feedback are related to a virtual body or action. Agency, ownership, and self-location provide separate but connected ways of describing this relation. Visuomotor congruence, spatial alignment, temporal synchrony, and plausible body mapping are therefore central design variables.

In XR rowing, these variables are unusually concrete. The user performs a real rowing-like movement on a physical ergometer. The system must decide whether the virtual representation should remain close to this physical setup or transform it into a more complete on-water rowing scene. Both strategies are defensible, but they support different forms of realism.

The ergometer-congruent mode prioritizes correspondence with the user’s real action. It is expected to support agency because the visible movement can follow the user’s own stroke closely. It may also support ownership because the virtual hands, body, handle, and seat can be arranged in a spatially plausible relation to the user’s felt body and apparatus. Its limitation is that it may feel less like the culturally recognizable image of rowing on water.

The boat-centred mode prioritizes sport depiction and environmental plausibility. It can show a rower in a shell, moving through water, using oars, and performing a visually recognizable rowing action. This may strengthen presence, self-location in the rowing scene, and the subjective meaning of the activity as rowing. Its limitation is that it may require a transformation between the felt ergometer handle path and the visible oar-based movement.

This thesis therefore does not assume that one representation is universally better. Instead, it expects that the two modes may produce different embodiment profiles. The ergometer-congruent mode may be stronger for direct agency and body correspondence. The boat-centred mode may be stronger for plausibility, sport realism, and the feeling of being situated in a rowing environment. Body ownership and self-location may depend on how participants integrate these cues.

The empirical comparison is built around this tension. By holding the general VR rowing context constant while varying the representation strategy, the study examines whether embodiment in XR rowing is more strongly supported by precise movement correspondence or by a more realistic sport depiction. The result should help clarify how embodiment concepts can inform the design of future XR exercise and sport systems.

Chapter 3

System and Prototype

Chapter 4

Study Design and Method

Chapter 5

Results

Chapter 6

Discussion

Chapter 7

Conclusion

Appendix A

Questionnaires

Appendix B

Study Materials

Appendix C

Additional Results

Appendix D

System Details

Appendix E

AI Assistance Statement

This appendix documents how AI tools (e.g., ChatGPT, Codex) were used in the research and writing of this thesis. It summarizes the nature of the assistance, how the outputs were verified, and the role of the author in reviewing and editing AI-generated content.